Digestive System: the GI map

Build this as we go. Every stop on the tube in order, and every organ that pours into it. Competencies 109 to 125. Name _____

THE TUBE: FOLLOW THE BITE

1. Mouth (oral cavity)
Epithelium: _____
Happens here: _____
Boundaries: _____

↓

2. Pharynx
Food uses: _____
Sealed off: _____
Gate out: _____

↓

3. Esophagus
Epithelium: _____
Outer layer: _____
Through the diaphragm at: _____
Gate out: _____

↓

4. Stomach
Regions in order: _____
Epithelium: _____
Extra muscle layer: _____
Folds: _____
Gate out: _____

↓

5. Small intestine
Duodenum, 25 cm: _____
Jejunum, 1 m: _____
Ileum, 2 m: _____
Surface: _____
Gate out: _____

↓

6. Large intestine
In order: _____
Only here: _____
Epithelium: _____
Job: _____

↓

7. Rectum and anal canal
Rectum: _____
Anal canal lining near the anus: _____
Two gates: _____

ACCESSORY ORGANS: WHAT POURS INTO THE TUBE, AND WHERE

Salivary glands, into the mouth
Parotid duct opens: _____
Submandibular duct opens: _____
Sublingual ducts open: _____

Teeth and tongue
Tongue muscles: _____
Papillae without taste buds: _____
Adult teeth: _____

Liver, into the duodenum by bile
Lobes: _____
Lobule: _____
Bile path: _____

Gallbladder
Regions: _____
Its duct: _____
Job: _____

Pancreas, into the duodenum
Regions: _____
Main duct joins the _____
Accessory duct opens at the _____
Acini (exocrine) and _____

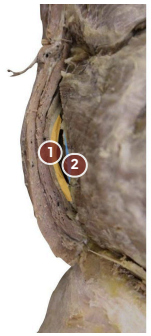
DRAW: THE WHOLE CANAL IN THE TORSO, DIAPHRAGM AT THE TOP, ACCESSORY ORGANS BESIDE IT, A RING AT EVERY SPHINCTER

Digestive System, Part I

Competencies 109 to 114, 120, 121: peritoneum and folds, retroperitoneal organs, the gut wall and its sphincters, mouth, salivary glands, tongue, esophagus.

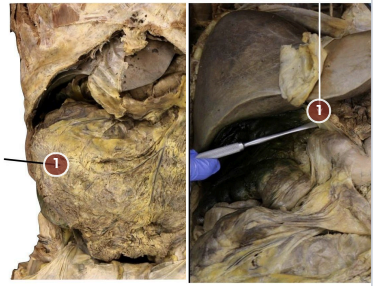
Name _____

A. Peritoneum and the omenta (cadaver) COMPETENCY 111



Cadaver, peritoneum: parietal and visceral layers

1 _____
2 _____



Cadaver, greater omentum | Cadaver, lesser or

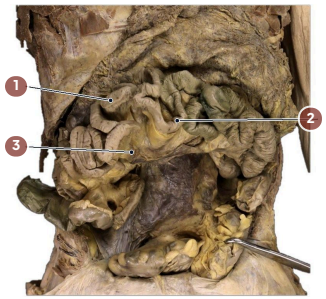
1 _____ 1 _____

FIVE FOLDS: WHAT EACH ONE CONNECTS

Greater omentum: _____
Lesser omentum: _____
Falciform ligament: _____
Mesentery: _____
Mesocolon: _____

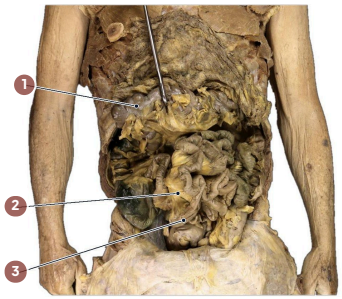
Extra notes and drawings

B. Mesentery, and what sits behind the peritoneum (cadaver) COMPETENCY 111, 112



Cadaver, mesentery of the small intestine

1 _____
2 _____
3 _____



Cadaver, transverse colon lifted

1 _____
2 _____
3 _____

FILL IN

Intraperitoneal: _____
Retroperitoneal: _____
Retroperitoneal organs: _____
On a mesocolon: _____

Extra notes and drawings

Digestive System, Part I

Competencies 109 to 114, 120, 121: peritoneum and folds, retroperitoneal organs, the gut wall and its sphincters, mouth, salivary glands, tongue, esophagus.

Name _____

C. The wall of the tube, and its gates (draw it) COMPETENCY 109, 110, 114

FOUR LAYERS, LUMEN OUTWARD

1. _____
2. _____
3. _____
4. _____

FILL IN

Submucosal plexus controls _____

Myenteric plexus controls _____

Stratified squamous lines the _____

Simple columnar lines the _____

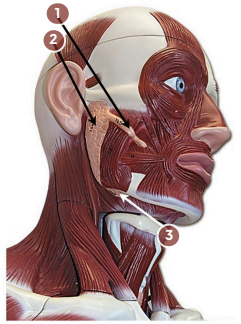
SIX SPHINCTERS, IN THE ORDER FOOD MEETS THEM

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

DRAW: A CROSS SECTION OF THE TUBE AS FOUR RINGS, WITH THE TWO PLEXUSES DOTTED IN

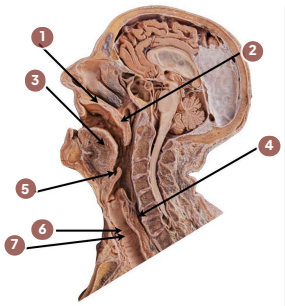
Extra notes and drawings

D. Mouth, salivary glands, tongue, esophagus COMPETENCY 113, 120, 121



Model, salivary glands

1. _____
2. _____
3. _____



Cadaver, midsagittal: palate, tongue, esophagus

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

FILL IN

Parotid duct: _____

Submandibular duct: _____

Sublingual ducts: _____

No taste buds: _____

Extrinsic muscles: _____

Intrinsic muscles: _____

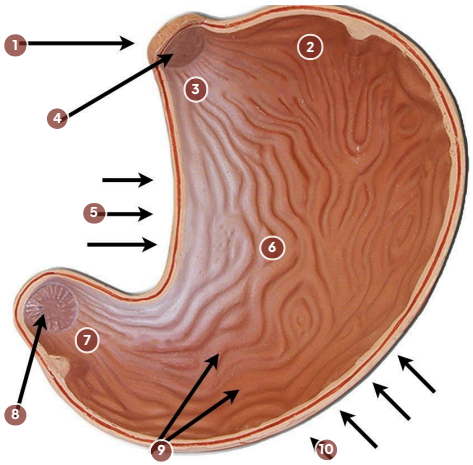
Esophageal hiatus: _____

Extra notes and drawings

Digestive System, Part II

Competencies 115 and 122 to 125: stomach, liver, gallbladder and bile path, pancreas, and the accessory organ disorders. Name _____

A. Stomach regions and features (model) COMPETENCY 115



Model, stomach, interior

1 _____ 6 _____
2 _____ 7 _____
3 _____ 8 _____
4 _____ 9 _____
5 _____ 10 _____

FILL IN

Muscle layers, inside out: _____

Extra layer: _____

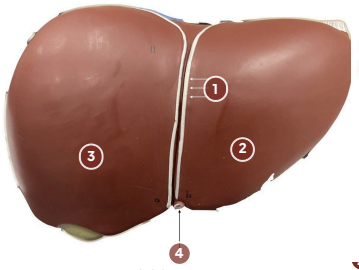
Lesser curvature faces _____

Greater curvature holds the _____

Rugae are folds of _____

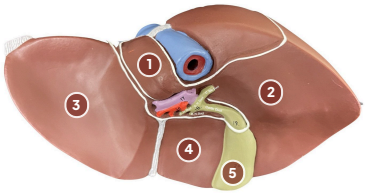
Extra notes and drawings

B. Liver lobes and ligaments (model) COMPETENCY 122



Model, liver, anterior

1 _____
2 _____
3 _____
4 _____



Model, liver, inferior

1 _____
2 _____
3 _____
4 _____
5 _____

FILL IN

From the front: _____

From below: _____

Falciform ligament reaches the _____

Ligamentum teres was the _____

Porta hepatis, in: _____

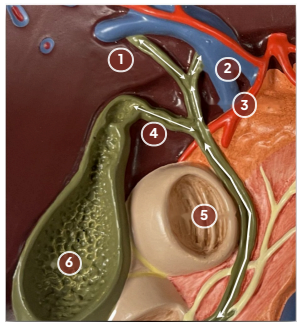
Porta hepatis, out: _____

Extra notes and drawings

Digestive System, Part II

Competencies 115 and 122 to 125: stomach, liver, gallbladder and bile path, pancreas, and the accessory organ disorders. Name _____

C. Bile path and the pancreas (model and cadaver) COMPETENCY 123, 124



Model, gallbladder and the biliary tree

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____

A DROP OF BILE, IN ORDER

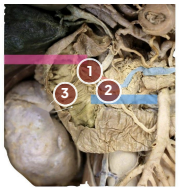
1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____

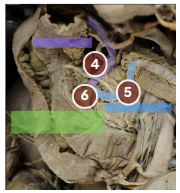
FILL IN

Joins the common bile duct: _____

Exocrine: _____

Endocrine: _____

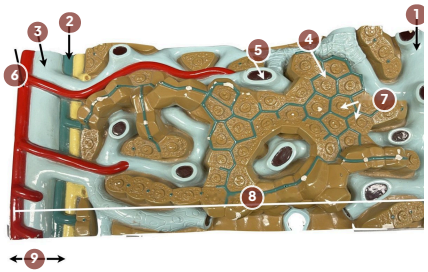




Cadaver, pancreatic and bile ducts entering the duodenum

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____

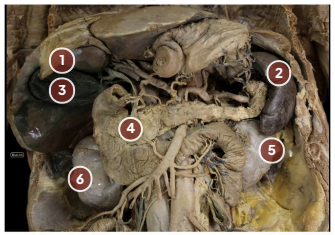
D. Liver lobule, and the organs in place (model and cadaver) COMPETENCY 122, 125



Model, liver lobule

1 _____
2 _____
3 _____
4 _____
5 _____

6 _____
7 _____
8 _____
9 _____



daver, major organs of the abdomen

1 _____
2 _____
3 _____

4 _____
5 _____
6 _____

FILL IN

Blood flows toward the _____

Bile flows toward the _____

Portal triad: _____

DISORDERS: NAME THE ORGAN

Gallstones, cholecystitis: _____

Hepatitis, cirrhosis: _____

Pancreatitis: _____

Mumps: _____

Stone at the ampulla blocks _____

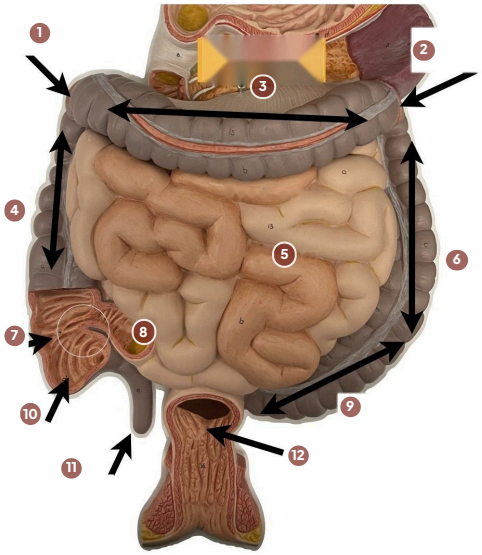
Extra notes and drawings

Extra notes and drawings

Digestive System, Part III

Competencies 116 to 119: small intestine, large intestine and rectum, the path of food, and the canal disorders. Name _____

A. Small and large intestine in place (model) COMPETENCY 116, 117



Model, small and large intestine

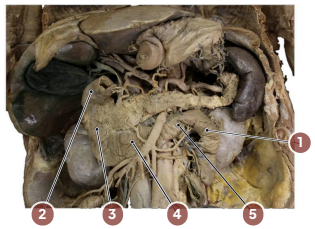
1 _____ 7 _____
2 _____ 8 _____
3 _____ 9 _____
4 _____ 10 _____
5 _____ 11 _____
6 _____ 12 _____

FILL IN

Duodenum: _____
Jejunum: _____
Ileum, the longest: _____
Food leaves the ileum into the _____
Four parts of the colon: _____

Extra notes and drawings

B. Duodenum (cadaver) and the colon surface (model) COMPETENCY 116, 117

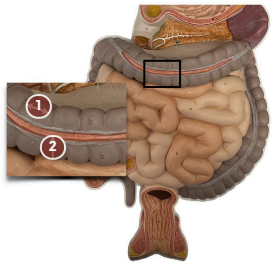


Cadaver, the four parts of the duodenum

1 _____
2 _____
3 _____
4 _____
5 _____

FILL IN

Folding, big to small: _____
Lacteal absorbs _____
Large intestine only: _____
Teniae coli are the _____
No haustra on the _____



Model, large intestine surface

1 _____
2 _____

Extra notes and drawings

Digestive System, Part III

Competencies 116 to 119: small intestine, large intestine and rectum, the path of food, and the canal disorders. Name _____

C. Trace a bite of food, mouth to anus COMPETENCY 118

IN ORDER, WITH A RING AT EACH SPHINCTER

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____

FILL IN

Not on the path: _____

Most absorption happens in the _____

Water is absorbed and feces form in the _____

Extra notes and drawings

D. Disorders: which structure does each one affect? COMPETENCY 119

DISORDER: STRUCTURE, WHAT CHANGES

GERD: _____

Hiatal hernia: _____

Peptic ulcer: _____

Appendicitis: _____

Diverticulosis: _____

IBD: _____

Hemorrhoids: _____

Colorectal cancer: _____

DRAW: THE FOUR WALL LAYERS, WITH AN ULCER ERODING FROM THE MUCOSA OUTWARD

Extra notes and drawings

Digestive System: Histology

The tube and the accessory organs under the microscope. Competencies 109, 113, 115 to 117, 121, 122, 124.

Name _____

A. Esophagus and stomach on the slide COMPETENCY 109, 113, 115



Slide, esophagus (right) beside the trachea

Slide photo to add

Slide: stomach wall, gastric pits and glands

FILL IN

Esophagus epithelium: _____

Esophagus outer layer: _____

Stomach surface epithelium: _____

Gastric pits lead to _____

Chief cells make _____

Parietal cells make _____

Stomach muscularis: _____

B. Small intestine and large intestine on the slide COMPETENCY 116, 117

Slide photo to add

Slide: small intestine, villi and crypts

Slide photo to add

Slide: colon, crypts with goblet cells, no villi

FILL IN

Villi are _____

Microvilli form the _____

Crypts (intestinal glands) hold _____

Duodenum only, in the submucosa: _____

Ileum only: _____

Colon mucosa: _____

Colon outer muscle: _____

Extra notes and drawings

Extra notes and drawings

Digestive System: Histology

The tube and the accessory organs under the microscope. Competencies 109, 113, 115 to 117, 121, 122, 124.

Name _____

C. Liver and pancreas on the slide COMPETENCY 122, 124

Slide photo to add

Slide: liver lobule, central vein and portal triad

Slide photo to add

Slide: pancreas, acini with a pale islet

FILL IN

Six-sided unit: _____

In the center: _____

At the corners: _____

Between the plates: _____

Blood flows _____

Bile flows _____

Pancreas: dark clusters are _____

Pale round patches are _____

Extra notes and drawings

D. Salivary gland on the slide, and the four-layer rule COMPETENCY 109, 121

Slide photo to add

Slide: salivary gland, serous and mucous cells

FILL IN

Serous cells make _____

Mucous cells make _____

Parotid is _____

Sublingual is _____

FOUR LAYERS, ANY SLIDE OF THE TUBE

1. _____
2. _____
3. _____
4. _____

DRAW: A WEDGE OF GUT WALL AS IT LOOKS ON A SLIDE, LUMEN AT THE BOTTOM, THE TWO PLEXUSES DOTTED IN

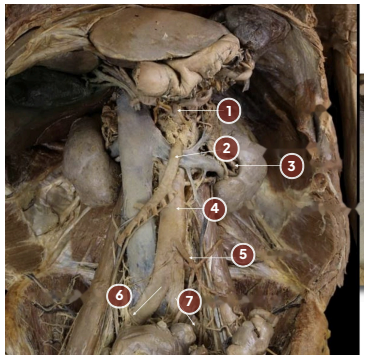
Extra notes and drawings

Blood Flow of the Abdomen and Lower Limb

Competencies 126 and 127: abdominal aorta and branches, the inferior vena cava, the hepatic portal system, pelvic and lower limb vessels.

Name _____

A. Abdominal aorta and its branches (cadaver and model) COMPETENCY 126



Cadaver, abdominal aorta and branches

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____

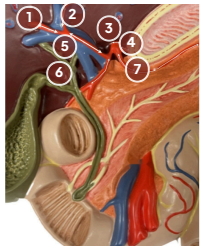
FILL IN

Unpaired, top to bottom: _____

Celiac trunk splits into _____

Paired branches: _____

Ends at L4 as the _____

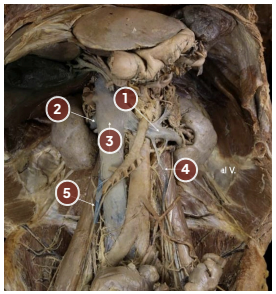


Model, celiac trunk and its branches

1 _____
2 _____
3 _____
4 _____
5 _____
6 _____
7 _____

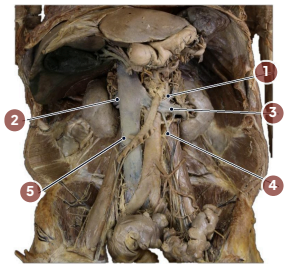
Extra notes and drawings

B. Inferior vena cava and its tributaries (cadaver) COMPETENCY 126



Cadaver, abdominal veins

1 _____
2 _____
3 _____
4 _____
5 _____



Cadaver, inferior vena cava and tributaries

1 _____
2 _____
3 _____
4 _____
5 _____

FILL IN

IVC forms at L5 from the _____

Through the diaphragm at _____

Right gonadal vein drains to _____

Left gonadal vein drains to _____

Hepatic veins enter _____

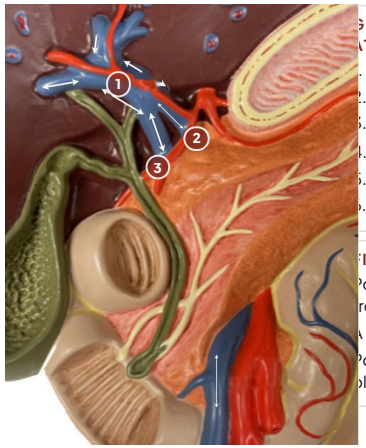
Extra notes and drawings

Blood Flow of the Abdomen and Lower Limb

Competencies 126 and 127: abdominal aorta and branches, the inferior vena cava, the hepatic portal system, pelvic and lower limb vessels.

Name _____

C. The hepatic portal system: the detour through the liver (model) COMPETENCY 126



Model, hepatic portal system

GLUCOSE, VILLUS TO RIGHT ATRIUM

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____

FILL IN

Portal vein forms from _____

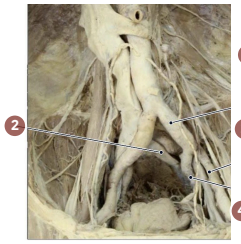
A portal system has _____

Portal vein carries blood _____

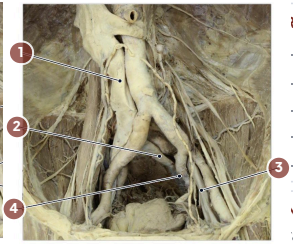
1 _____
2 _____
3 _____

Extra notes and drawings

D. Pelvis and lower limb: one artery, renamed at each landmark COMPETENCY 127



Cadaver, iliac arteries



Cadaver, iliac veins

1 _____
2 _____
3 _____
4 _____

1 _____
2 _____
3 _____
4 _____

RY TO THE FOOT, IN ORDER

_____
_____
_____
_____
_____

N

al triangle, lateral to medial: _____

Superficial vein up the medial leg: _____

Sciatic nerve splits into _____

Extra notes and drawings